Evidencing preferential attachment in dependency network evolution

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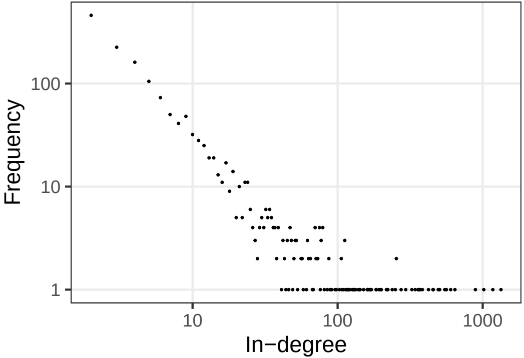
Introduction

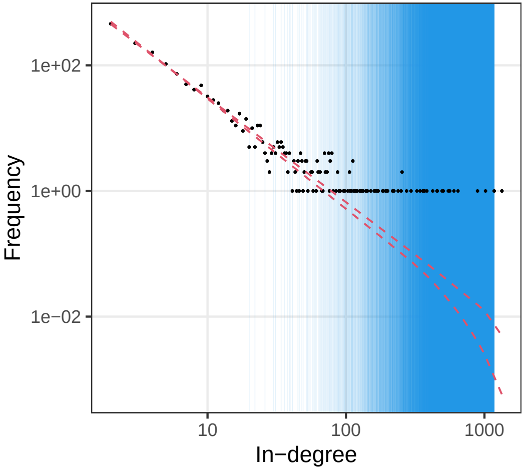
- Interest in degree distribution
- Power law $\rightarrow$ Scale-freeness
- $\rightarrow$ Preferential attachment
  1. New node joins & brings new edges
  2. Node i selected with weight $g(d_i)$
  3. Barabási and Albert (1999):
     $g(d_i) = d_i$



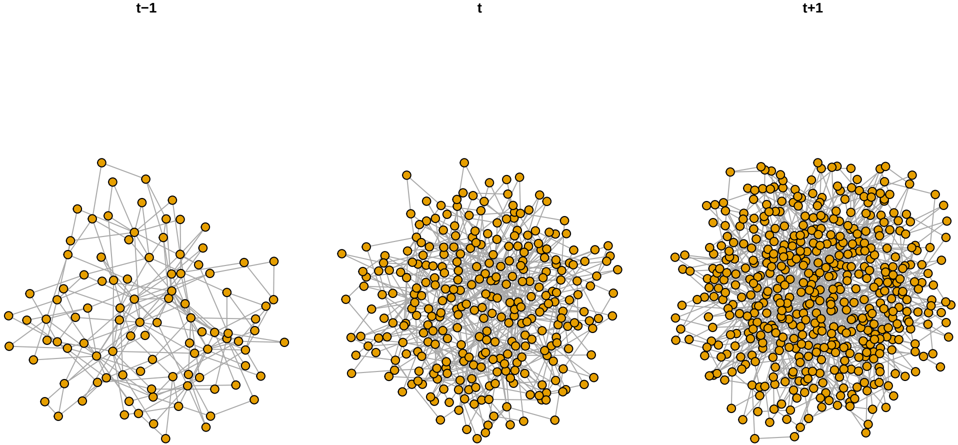
A few issues

1. General case $\nrightarrow$ power law
   - $g(d_i) = d_i^\alpha$ (Krapivsky and Redner 2001)
2. Other models $\rightarrow$ scale-freeness
   - Generalised random graphs (Hofstad 2016)
3. Debate on ubiquity
   - Broido and Clauset (2019): rare in real networks
   - Voitalov et al. (2019): rebuttal
   - Lee, Eastoe, and Farrell (2024): partial scale-freeness

- Main cause: only snapshots available



Taking one step back



Goals

If we have the evolution data:

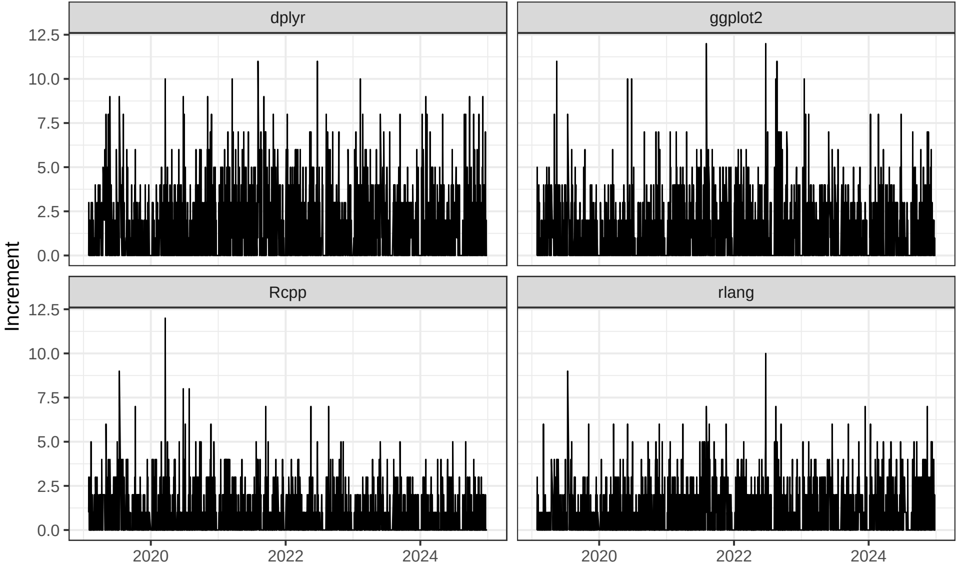
1. Can we evidence the preferential attachment?
2. How do we model the data?
3. What is the precise form of the weight function?
    - $g(d_i) = d_i^\alpha$ (including $\alpha = 1$)?
    - $g(d_i) =$ a more general form?

Data: Daily increments of R packages on CRAN

- Imports for one day

```
## # A tibble: 10 x 3
##    from            to      add
##    <chr>           <chr>   <lgl>
##  1 glmmsr          methods FALSE
##  2 GpGp            FNN     TRUE
##  3 pmxTools        stats   FALSE
##  4 ceterisParibus  knitr   TRUE
##  5 pmxTools        xpose   FALSE
##  6 mRchmadness     shiny   FALSE
##  7 glmmsr          lme4    FALSE
##  8 mRchmadness     rvest   FALSE
##  9 xpose           dplyr   FALSE
## 10 xpose           utils   FALSE
```

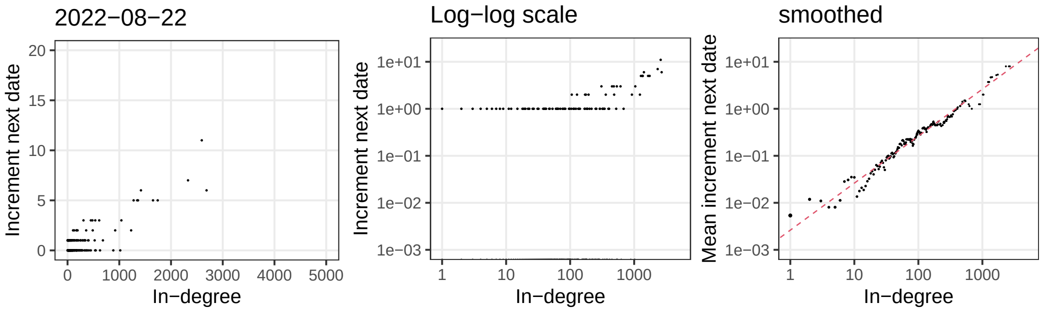
Aggregating once



Aggregating twice

```
## # A tibble: 355,785 x 4
##    `previous date` indegree increment count
##    <date>             <dbl>     <dbl> <dbl>
##  1 2019-01-29             0         0  9315
##  2 2019-01-29             0         1     1
##  3 2019-01-29             1         0  1306
##  4 2019-01-29             2         0   459
##  5 2019-01-29             3         0   226
##  6 2019-01-29             4         0   161
##  7 2019-01-29             5         0   103
##  8 2019-01-29             6         0    75
##  9 2019-01-29             7         0    49
## 10 2019-01-29             8         0    40
## # i 355,775 more rows
```

Scatter plot



Tweaking the original model

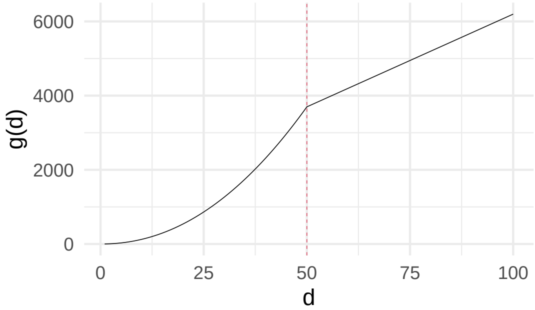
- Y_i: number of new edges / increments for node i
- 1 new edge, $(Y_1, Y_2, \ldots) \sim \text{Multinomial}\left(\frac{d_1^\alpha}{\sum_j d_j^\alpha}, \frac{d_2^\alpha}{\sum_j d_j^\alpha}, \ldots\right)$
  - Essentially the **statistical** model based on Barabási and Albert (1999) & Krapivsky and Redner (2001)
- $M \sim \text{Poisson}(m)$ new edges,
  $(Y_1, Y_2, \ldots)|M = m \sim \text{Multinomial}\left(\frac{d_1^\alpha}{\sum_j d_j^\alpha}, \frac{d_2^\alpha}{\sum_j d_j^\alpha}, \ldots\right)$
  - $(Y_1, Y_2, \ldots) \sim$ Independent Poissons with means $\left(\frac{m\ d_1^\alpha}{\sum_j d_j^\alpha}, \frac{m\ d_2^\alpha}{\sum_j d_j^\alpha}, \ldots\right)$
  - $\log E(Y_i) = \alpha \log d_i + c$

Tail behaviour poorly captured when $g(d) = d^\alpha$

- Krapivsky and Redner (2001)
  - $\alpha < 1$ (sublinear): Weibull tail (long-tailed but light-tailed)
  - $\alpha > 1$ (superlinear): Degeneracy
- More general result by Rudas, Tóth, and Valkó (2007)
- Empirically
  - Subtle tail behaviour (Lee, Eastoe, and Farrell 2024) in real data
  - Heavy-tailed, but not as heavy as implied by when $\alpha = 1$

Recent findings

- $g(d) = \begin{cases} d^{\alpha}, & d \leq \gamma, \\ \gamma^{\alpha} + \beta(d - \gamma), & d \geq \gamma \end{cases}$
- Sub/super-linear up to threshold γ, then linear above
- Flexible heavy-tail behaviour



Actual modelling

1. Regress increment on (in-)degree
   - $Y_i \sim \text{Poisson}\left(\frac{m\, g(d_i)}{\sum_j g(d_j)}\right)$
2. Choice of $g(d)$
   - $g(d) = \begin{cases} d^\alpha + \delta, & d \leq \gamma, \\ \gamma^\alpha + \beta(d-\gamma) + \delta, & d \geq \gamma \end{cases}$ (piecewise function)
   - $g(d) = d^\alpha + \delta$ (power function, for benchmark)
3. Infer (α, δ) (and (β, γ) if necessary)
   - Results of α most important

Back to CRAN packages

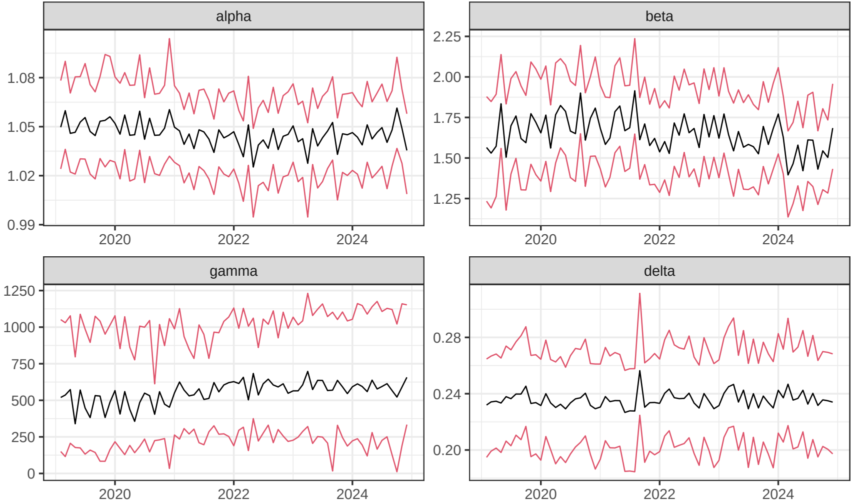
rstan: R Interface to Stan

User-facing R functions are provided to parse, compile, test, estimate, and analyze Stan models by accessing the header-only Stan library provided by the 'StanHeaders' package. The Stan project develops a probabilistic programming language that implements full Bayesian statistical inference via Markov Chain Monte Carlo, rough Bayesian inference via 'variational' approximation, and (optionally penalized) maximum likelihood estimation via optimization. In all three cases, automatic differentiation is used to quickly and accurately evaluate gradients without burdening the user with the need to derive the partial derivatives.

Version:	2.32.7
Depends:	R (≥ 3.4.0), StanHeaders (≥ 2.32.0)
Imports:	methods, stats4, inline (≥ 0.3.19), gridExtra (≥ 2.3), Rcpp (≥ 1.0.7), RcppParallel (≥ 5.1.4), loo (≥ 2.4.1), pkgbuild (≥ 1.2.0), QuickJSR, ggplot2 (≥ 3.3.5)
LinkingTo:	Rcpp (≥ 1.0.7), RcppEigen (≥ 0.3.4.0.0), BH (≥ 1.75.0-0), StanHeaders (≥ 2.32.0), RcppParallel (≥ 5.1.4)
Suggests:	testthat (≥ 3.0.4), parallel, KernSmooth, shinystan, bayesplot, rmarkdown, rstantools, rstudioapi, Matrix, knitr, coda, V8

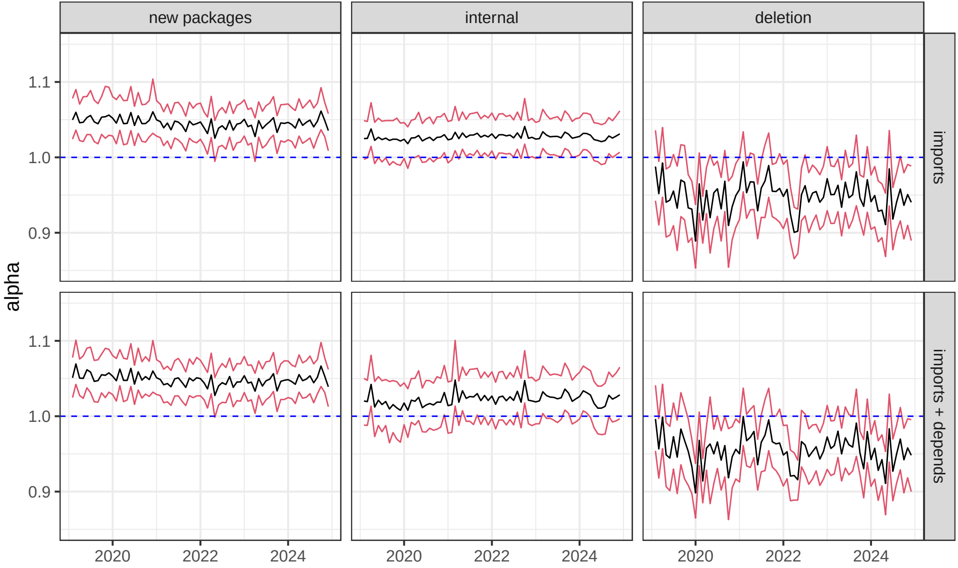
- Type of dependencies: `Imports` only, as well as `Imports` + `Depends`
- New packages, or internal, or deletion
- Increments of all days in each month bundled
- A set of parameters for each month – $(\{\alpha_t\}, \{\beta_t\}, \{\gamma_t\}, \{\delta_t\})$

Piecewise function, `Imports`, new packages



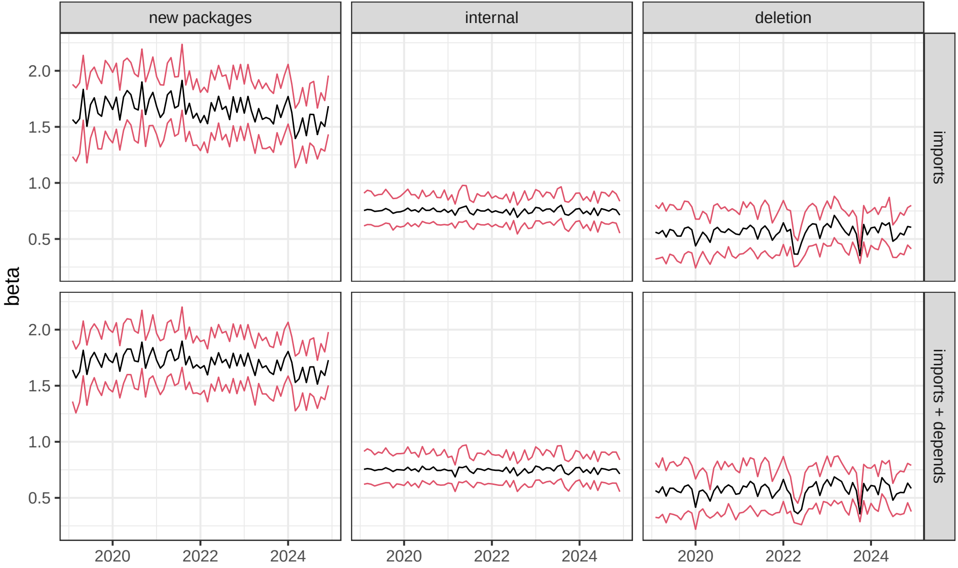
- Very similar results for α & δ using power function

α

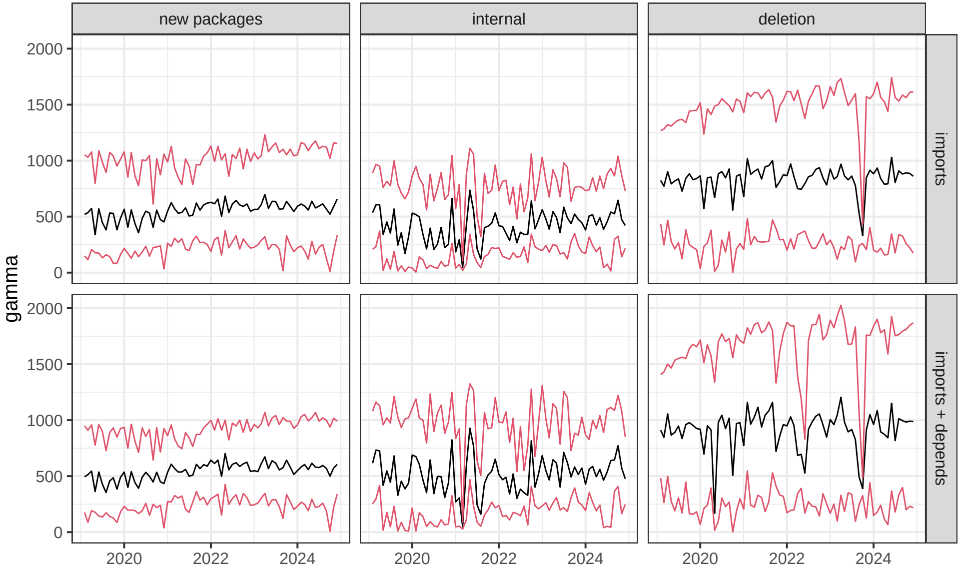


new packages
internal
deletion
imports
imports + depends
alpha
1.1
1.0
0.9
2020
2022
2024

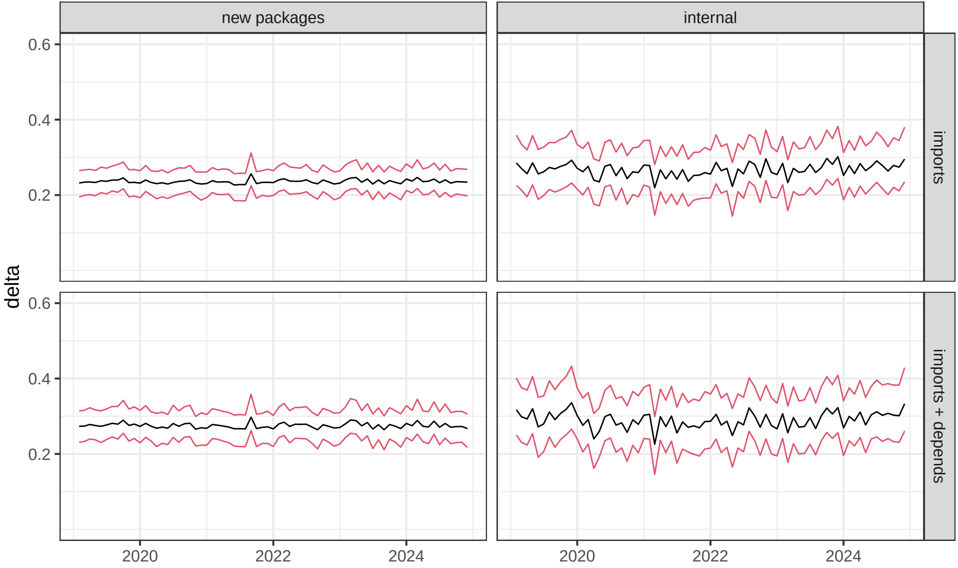
β


new packages
internal
deletion
imports
imports + depends
beta
2.0
1.5
1.0
0.5
2020
2022
2024

γ


new packages
internal
deletion
imports
imports + depends
gamma
2000
1500
1000
500
0
2020
2022
2024

δ


new packages
internal
imports
imports + depends
delta
0.6
0.4
0.2
2020
2022
2024

Overall

- Parameters stable over the observation period
- Additing edges (new packages or internal):
  - **Partial superlinear preferential attachment (up to γ)**
- Deleting edges:
  - **Sublinear preferential detachment without zero-appeal (δ)**
- Not much difference between `Imports` and `Imports` + `Depends`

Summary

- Preferential attachment evidenced from increments of network evolution
  - exploratory visualisations
  - Poisson regression model
- Piecewise weight function of (in-)degree
  - accommodates flexible heavy tail (of the degree distribution)
  - reveals partial super-/sub-linear preferential attachment
- Model applied to CRAN package dependencies successfully
- Next step: Model selection between power & piecewise functions

Bibliography

Barabási, Albert-László, and Réka Albert. 1999. "Emergence of Scaling in Random Networks." *Science* 286 (5439): 509–12. https://doi.org/10.1126/science.286.5439.509.

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Rudas, A, B Tóth, and B Valkó. 2007. "Random Trees and General Branching